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5(a)(i)	A calculation in which: • substitute values correctly into the equation (1) • correct answer with units (1)	<u>Example of calculation</u> $V = \frac{4 \times \pi \times 50 \times 50 \times 50}{3}$ volume = 523 599 / 5.2 x10⁵ nm³ CE applies if 100 has been used in calculation instead of 50 ALLOW correct conversions with different units No working: e.g. 523 599 / 5.2 x10⁵ nm³ / 523 333 nm³ gains 2 marks e. g. 4 186 667 / 4.2 x10⁶ nm³ gains 1 mark	(2)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that includes the following points: • because {hydrophobic / non-polar} tails {move away from / repelled by} the {aqueous environment / water} (1) • {hydrophilic / polar} heads {interact with / associate / dissolve in} the {aqueous environment / water} (1)	NB if no other marks awarded, allow 'hydrophilic heads face water and hydrophobic tails face away from water' for 1 mark	(2)

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5(b)(i)	The only correct answer is B. <i>A is incorrect because the liposome has no protein and cannot generate ATP.</i> <i>C is incorrect because liposomes cannot take up molecules by endocytosis.</i> <i>D is incorrect because water only moves by osmosis.</i>	(1)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	A description that includes the following points: • increase in temperature increases membrane permeability (1) • increase in cholesterol decreases membrane permeability (1) • cholesterol has a greater effect on membrane permeability at higher temperatures (1)	ACCEPT converse throughout	(3)

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5(b)(iii)	An explanation that includes the following points: • because an increase in temperature increases the movement of phospholipids (1) • because the cholesterol decreases fluidity (of the membrane) (1)	ACCEPT fills the gaps between the phospholipid tails / acts as a barrier / reduces movement of phospholipids	(2)